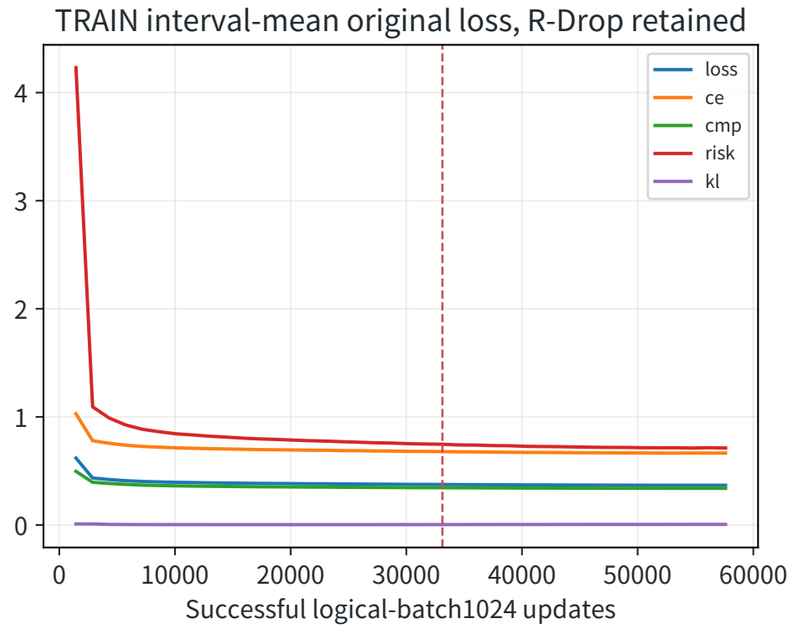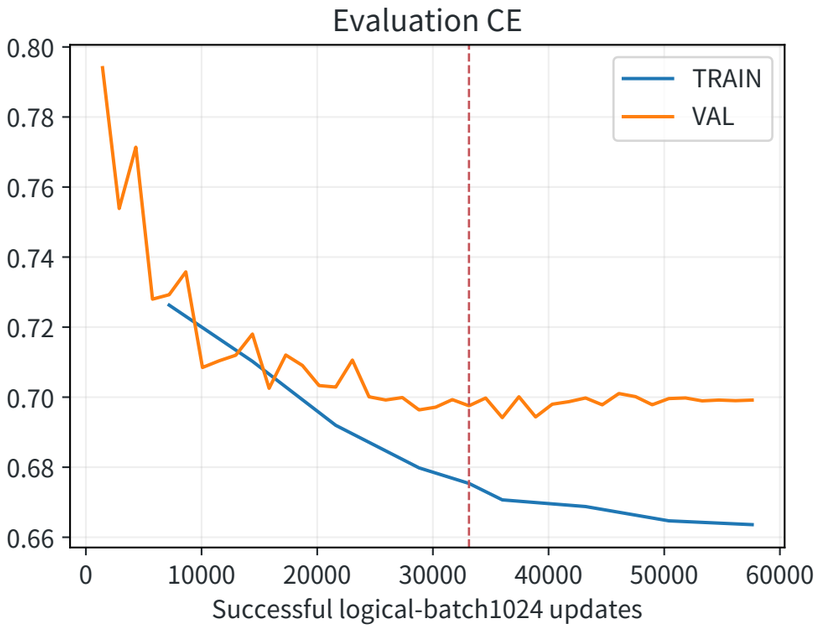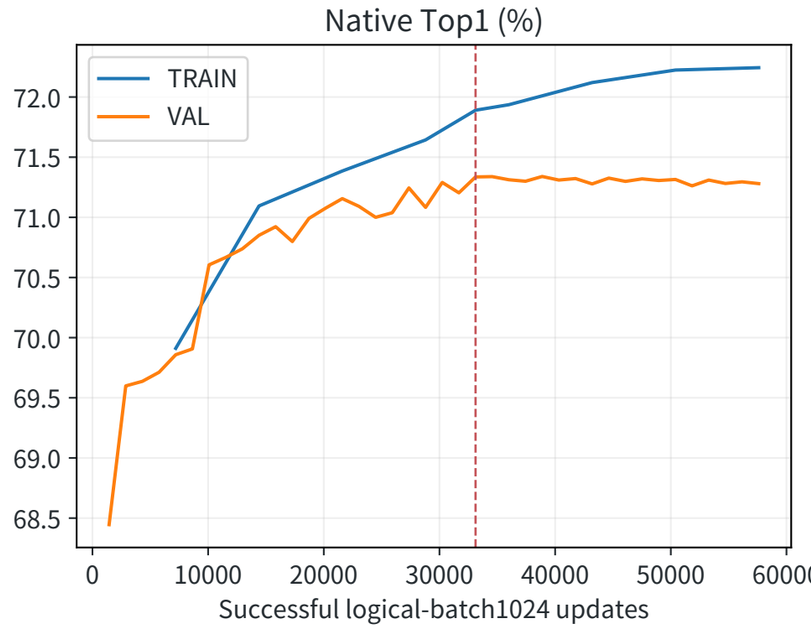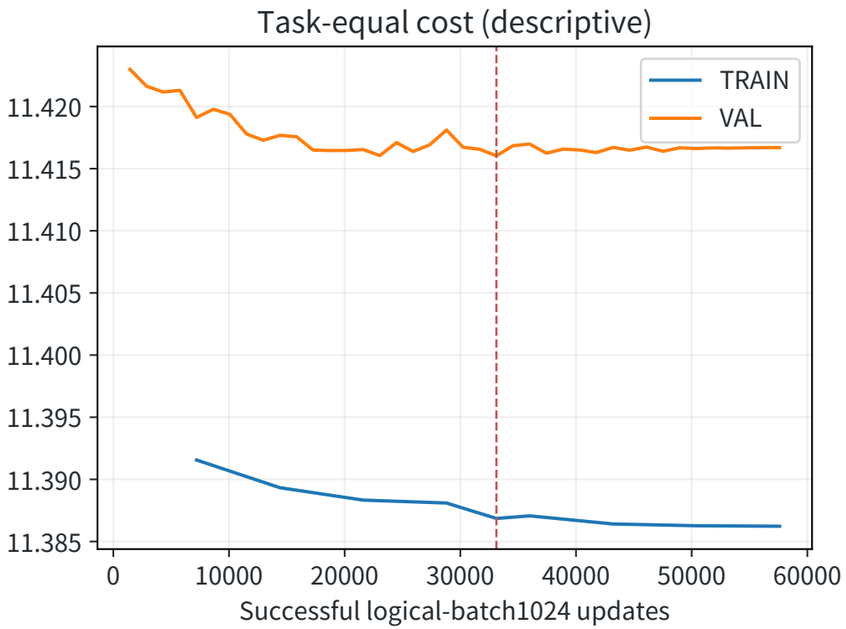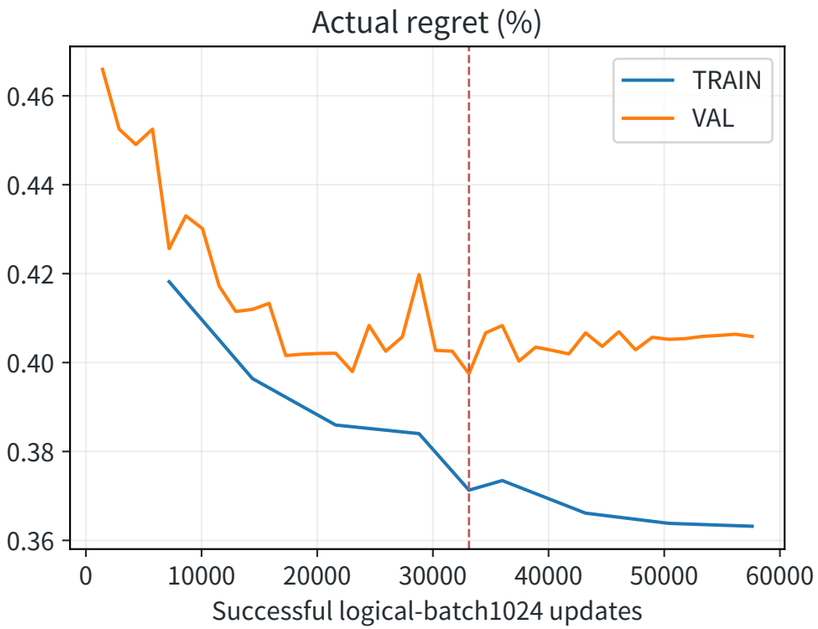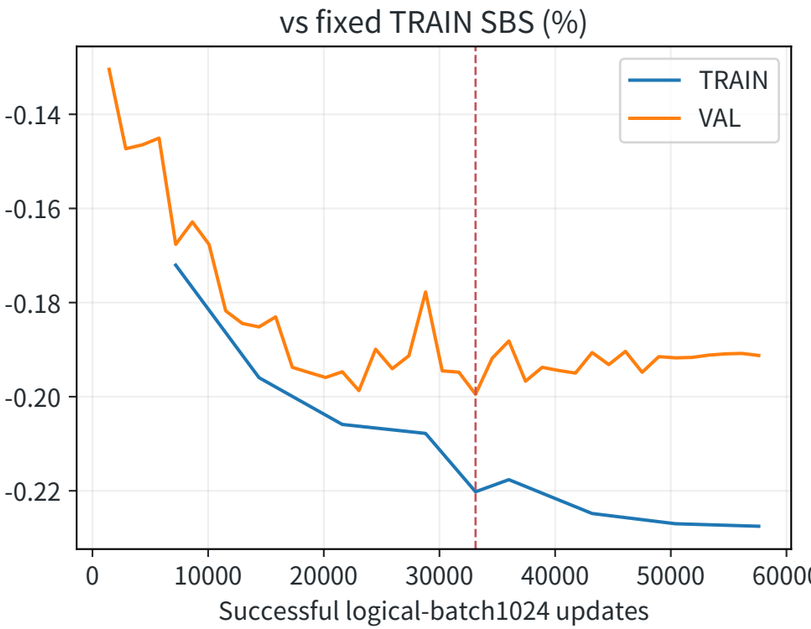R106/A_frozen | original R77; online native; one VAL checkpoint